INCH-POUND

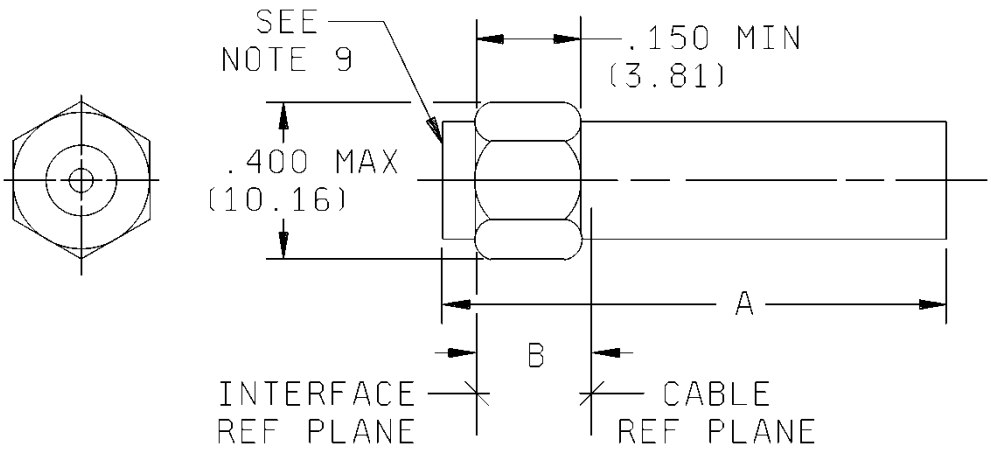
MIL-PRF-39012/79G
w/AMENDMENT 2
14 October 2016
SUPERSEDING
MIL-PRF-39012/79G
w/AMENDMENT 1
4 March 2010

PERFORMANCE SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, COAXIAL, RADIO FREQUENCY, SERIES SMA (CABLED) PIN CONTACT, CLASS 2, SEMIRIGID CABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39012.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Dimension .400 (10.16) in the largest overall envelope of the connector.
4. Wrench flats are to accommodate standard wrench opening in accordance with FED-STD-H28, appendix 10.
5. Dimension A defines the length of the connector when assembled to the cable.
6. When applicable, three holes .016 (0.41 mm) minimum diameter, equally spaced, are required for safety wiring after mating. Location on the coupling nut is optional.
7. All undimensioned pictorial configurations are for reference purposes only.
8. All threads shall be in accordance with FED-STD-H28.
9. Series SMA, pin contact interface in accordance with MIL-STD-348.

FIGURE 1. General configuration.

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TABLE I. Dash numbers and dimensions.

Dash no. M39012/79-	Applicable cable 1/ M17/	Dimension A (millimeters) 2/		Dimension B (millimeters)
		Maximum before assembly	Maximum after assembly	
CATEGORY A – FIELD SERVICEABLE (NO SPECIAL TOOLS REQUIRED) 3/ 4/ 5/				
3009 3109 6/ 4009 4106 6/	133-RG405 *	.513 (13.03 mm)	.460 (11.68 mm)	Not applicable
3010 3110 6/ 4010 4110 6/	130-RG402 *			
CATEGORY E – FIELD REPLACEABLE (STANDARD ASSEMBLY TOOL KIT) 4/ 5/ 7/ 8/				
3007 3/ 3107 3/ 6/ 4007 3/ 4107 3/ 6/	133-RG405 *	Not applicable	.828 (21.03 mm)	Not applicable 9/
3008 3/ 3108 3/ 6/ 4008 3/ 4108 3/ 8/	130-RG402 *			
3011 3/ 10/ 3111 3 / 6/ 10/ 4011 3/ 10/ 4111 3/ 6/ 10/	133-RG405 *		.460 (11.68 mm)	.215 (5.46 mm) ±.010 (0.25 mm)
3012 3/ 10/ 3112 3/ 6/ 10/ 4012 3/ 10/ 4112 3/ 6/ 10/	130-RG402 *			

See notes at end of table.

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TABLE I. Dash numbers and dimensions – Continued.

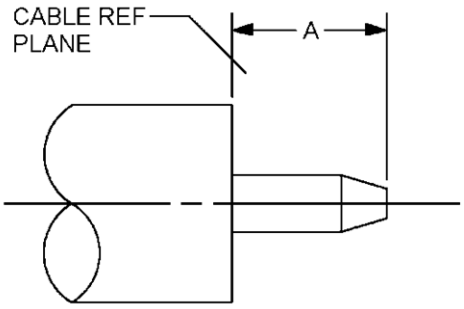
Dash number M39012/79-	Applicable cable 1/ M17/	Tool number M22520/	Positioning dies M22520/	Locator pins 7/ M22520/36	Crimp dies M22520/5	Dimension A		Dimension B
						Before crimping 2/	After crimping	
CATEGORY F – FIELD REPLACEABLE (MIL-DTL-22520 CRIMP TOOL) 3/ 4/								
3207 12/ 3307 6/ 12/ 4207 12/ 4307 6/ 12/	133-RG405 *	36-01	36-02	-04, -18 or -20	--------	.690 (17.53 mm) maximum	.568 (16.43 mm) maximum	Not applicable 9/
3208 12/ 3308 6/ 12/ 4208 12/ 4308 6/ 12/	130-RG402 *		36-03					
3210 3310 6/ 4210 4310 6/	130-RG402 *	5-01	--------	--------	-05 or -41 cavity B	.667 (16.94 mm) maximum	.667 (16.94 mm) maximum	
3211 10/ 12/ 3311 6/ 10/ 12/ 4211 10/ 12/ 4311 6/ 10/ 12/	133-RG405 *	36-01	36-02	-20	---------	.513 (13.03 mm) maximum	.460 (11.68 mm) maximum	To be determined
3212 10/ 12/ 3312 6/ 10/ 12/ 4212 10/ 12/ 4312 6/ 10/ 12/	130-RG402 *		36-03					
3213 3313 6/ 4213 4313 6/	133-RG405 *	5-01	---------	----------	-05 or -41 cavity B	.468 (11.89 mm) maximum	.468 (11.89 mm) maximum	Not applicable

See notes at end of the table.

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TABLE I. Dash numbers and dimensions – Continued.

* Cable to be used when performing test requiring cable. All cable dash numbers in accordance with MIL-DTL-17 specification sheet accommodate this connector.
1/ MIL-DTL-17 cables are specified by the basic number. The latest version of each cable shall be Applicable
2/ The dimension A, before crimping, determines the closeness of a bend in the cable to the rear of the connector.
3/ These connectors have captivated center contacts.
4/ For logistics purposes, only connectors with safety wire holes will be stocked.
5/ Coupling nuts shall be corrosion-resistant steel with a passivated finish in accordance with SAE-AMS2700, type 2. (Applies to "-3XXX" series connectors only).
6/ No safety wire holes.
7/ Kit number – Omni Spectra T-200, Amphenol 901-2500 or equivalent.
8/ All corrosion-resistant steel-bodied connectors shall be gold plated in accordance with ASTM B488, type II, grade C, class 1.27, at least in the area of solder attachment.
9/ This dimension is furnished in manufacturers assembly instructions.
10/ Gasket material shall be in accordance with SAE-AMS7276, SAE-AMS7259, SAE-AMS3216 and SAE-AMS3218.
11/ The locators required shall be indicated in the assemble instructions.
12/ Not for Army use, for OEM use only.



Cable *	Dimension "A"	
	Minimum	Maximum
M17/130-RG402	.080 (2.03 mm)	.090 (2.29 mm)
M17/133-RG405	.065 (1.65 mm)	.075 (1.91 mm)

* Dimensions apply to all cables on the below specification sheets.

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Tapering of center conductor is optional.

FIGURE 2. Cable stripping dimensions for category E and category F connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.
Frequency range: 0 to 18 GHz.
Voltage rating: The voltage rating shall be as specified in table II.

TABLE II. Voltage rating.

Cables	Voltage max. (sea level) V rms	Voltage max. at 70,000 ft. V rms
M17/133-RG405 *	335	85
M17/130-RG402 *	500	125

* These voltages apply to all cable Part or Identifying Numbers (PIN) on these specification sheets even if not listed.

Temperature rating: -65°C to +165°C.

Category F: Connectors using semirigid cables with standard stripping dimensions and using standard military assembly tools. The method of assembly of the connector to the cable shall be solderless.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and MIL-STD-348.
Force to engage and disengage:
Longitudinal force: Not applicable.
Torque: 2 inch-pounds, maximum.
Coupling proof torque: 15 inch-pounds, minimum.
Safety wire hole tearout: 15 pounds, minimum.

Inspection conditions: For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds.

Mating characteristics: See MIL-STD-348.
Hermetic seal: Not applicable.
Leakage (pressurized connectors): Not applicable.

Insulation resistance: In accordance with MIL-STD-202-302, test condition B, 5,000 megohms minimum.

Center contact retention:
Axial force: 6 pounds, minimum axial force. Applicable to captivated center contacts only.
Radial torque: Not applicable.

Salt atmosphere (corrosion): In accordance with MIL-STD-202-101, test condition B.

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Voltage standing wave ratio (VSWR): From 500 MHz to 18 GHz, or approximately 80 percent of the upper cutoff frequency of the cable, whichever is lower.

Cable group	VSWR	
	Captive	Non-captive
M17/133-RG405*	1.07 +0.01F (F in GHz)	1.07 +0.008F (F in GHz)
M17/130-RG402*	1.05 +0.01F (F in GHz)	1.05 +0.008F (F in GHz)

* VSWR values apply to all cable part numbers on these specification sheets even if not listed.

Swept frequency VSWR test setup:

Item 6: VSWR shall be less than 1.025 +.002F (F in GHz).

Item 16: VSWR shall be less than 1.025 +.002F (F in GHz).

Second step of VSWR checkout procedure – VSWR shall be less than 1.080 +.005F (F in GHz).

Group B inspection: Use step 5, long cable method.

Qualification and group C inspection: Use step 5, long cable method.

Connector durability: 500 cycles minimum at 12 cycles per minute maximum. The connector shall meet mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms, maximum.

	Initial	After environment
Center contact:	3.0	4.0
Outer contact:	2.0	Not applicable
Outer conductor to body:	.5	Not applicable

NOTE: Five milliohms are permissible on passivated steel bodied connectors (outer conductor to body only).

Dielectric withstanding voltage at sea level: In accordance with MIL-STD-202-301.

Cable group	V rms
M17/133-RG405 *	1,000
M17/130-RG402 *	1,500

* Values apply to all cable PIN's on these specification sheets even if not listed.

Vibration, high frequency: In accordance with MIL-STD-202-204, test condition D, except the method of mounting shall be approved by the qualifying activity.

Shock: In accordance with MIL-STD-202-213, test condition I.

Thermal shock: In accordance with MIL-STD-202-107, test condition B, except test high temperature shall be +115°C.

Moisture resistance: In accordance with MIL-STD-202-106. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Altitude: 70,000 feet.

Cable group	Volts, min.
M17/133-RG405 *	250
M17/130-RG402 *	375

* Values apply to all cable PIN's on these specification sheets even if not listed.

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RF high potential withstanding voltage:
Voltage and frequency: Tested at a frequency from 5 to 7.5 MHz.
Leakage current: Not applicable.

Cable group	Volts, min.
M17/133-RG405 *	670
M17/130-RG402 *	1,000

* Values apply to all cable PIN's on these specification sheets even if not listed.

Cable retention force: The cable retention force shall be as specified in table III.

TABLE III. Cable retention force.

Cable *	Pounds, minimum	Torque 1/ Inch-ounces
M17/133-RG405	30	16
M17/130-RG402	60	55

* Values apply to all cable PIN's on the below specification sheets even if not listed.
1/ Torque to be applied 4 inches maximum from the end of the connector. Reverse bend not applicable.

Coupling mechanism retention force: 60 pounds, minimum.
RF leakage: -90 dB minimum, tested at frequency between 2 and 3 GHz.
RF insertion loss: .03 $\sqrt{F(GH_z)}$ dB maximum tested at 9 to 12.4 GHz.

Supplemental test requirements for qualification and group C inspection for category F connectors.

Six additional connectors shall be selected from the production lot and terminated to produce three cable leads, twelve inches minimum, sixteen inches maximum. The three assemblies shall be subjected to the following tests in sequence: VSWR, thermal shock, VSWR, and cable retention.

The following exceptions to the requirements apply to this supplemental test only:

VSWR of the cable assembly: From 0.5 to 18 GHz.

Test cable assembly	VSWR (initial)	VSWR (after thermal shock)
M17/133-RG405 *	1.15 +.023F (F in GHz)	1.15 +.028F (F in GHz)
M17/130-RG402 *	1.10 +.023F (F in GHz)	1.10 +.025F (F in GHz)

* Values apply to all cable PIN's on these specification sheets even if not listed.

Thermal shock: 10 cycles.

Group qualification: See table IV.
PIN: M39012/79 (dash number in table I or "B" number in table V).
PIN cross-reference: See table VI.
Maintenance replacements for category B: See table VII.

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TABLE IV. Group qualification and retention testing.

Group	Submission and qualification of any of the following dash numbers 1/ 2/ 3/ M39012/	Qualifies the following dash numbers M39012/
I	79-*02	79-*01 79-*02
II	79-*04	79-*03 79-*04
III	79-*08 79-*12	79-*07 79-*08 79-*11 79-*12
IV	79-*10	79-*09 79-*10
V	79-*208 79-*212	79-*207 79-*208 79-*211 79-*212 79-*307 79-*308 79-*311 79-*312
VI	79-*210	79-*210 79-*213 79-*310 79-*313

1/ Individual connectors other than listed are self qualifying only.
2/ Qualification of connectors qualifies connectors of the same material only.
3/ Connectors qualified with safety wire holes automatically qualifies connectors without safety wire holes.
* Denotes body material and connector requirement.

NOTE: If a connector manufacturer produces a connector which meets all the requirements for two or more connector PINs (within the same series), the manufacturer may receive qualification approval for two or more connector PINs by qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate PIN. For group qualification, the connectors must be of similar design.

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TABLE V. Category B – Non-field replaceable (special tools may be required). 1/

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M39012/79B 2/ 3/ 4/	Applicable cable 5/	Dimension A maximum #
3001 6/ 3101 6/ 7/	M17/133-RG405 *	.828 (21.03 mm)
3002 6/ 3102 6/ 7/	M17/130-RG402 *	
3003 8/ 3103 7/ 8/ 4003 8/ 4103 7/ 8/	M17/133-RG405 *	
3004 8/ 3104 7/ 8/ 4004 8/ 4104 7/ 8/	M17/130-RG402 *	

* Cable to be used when performing tests requiring cable. All cable PIN's on this specification sheet shall accommodate this connector.
\# Dimensions are in inches. Metric equivalents are given for information only.
1/ Dimension B shown on figure 1 is not applicable to these connectors.
2/ Coupling nuts shall be corrosion-resistant steel with a passivated finish in accordance with SAE-AMS2700, type 2. (Applies to "-3XXX" series connectors only).
3/ For logistics purposes, only connectors with safety wire holes will be stocked.
4/ All corrosion resistant steel bodied connectors which are designed to be assembled to the outer conductor using solder shall be gold plated in accordance with ASTM B488, type II, code C, class 1.27.
5/ MIL-DTL-17 cables are specified by the basic number. The latest version of each cable shall be applicable.
6/ In active for new design.
7/ No safety wire holes.
8/ These connectors have captivated center contacts.

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TABLE VI. Cross reference of PIN's.

Replacement PIN M39012/79	Substitute for PIN or type designation M39012/79
B3001	-3001
B3103	-3101
B3002	-3002
B3102	-3102
B3003	-3801, -3803, -3003
B3103	-3103
B3004	-3802, -3804, -3004
B3104	-3104
B4003	-4801, -4803, -4001, -4003
B4103	-4101, -4103
B4004	-4802, -4804, -4002, -4004
B4104	-4102, -4104
-3007	-3807
-3008	-3808
-4007	-4807
-4008	-4808
-3007	-3005
-3107	-3105
-3008	-3006
-3011	84149SSGA
-3012	84149SSG
-3108	-3106
-4007	-4005
-4107	-4105
-4008	-4006
-4108	-4106

TABLE VII. Maintenance replacements for category B.

Category B number 1/	Category E dash number
B3001	3007
B3101	3107
B3002	3008
B3102	3108
B3003	3011
B3103	3111
B4003	4011
B4103	4111
B3004	3012
B3104	3112
B4004	4012
B4104	4112

1/ Category B connectors are for original installation only. They will not be stocked or procured by the government.

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Referenced documents. In addition to MIL-PRF-39012, this document references the following:

FED-STD-H28	MIL-STD-202-204	MIL-STD-348	SAE-AMS2700
MIL-STD-202-101	MIL-STD-202-213	MIL-DTL-17	SAE-AMS7276
MIL-STD-202-106	MIL-STD-202-301	MIL-DTL-22520	SAE-AMS7259
MIL-STD-202-107	MIL-STD-202-302	ASTM B488	SAE-AMS3216
			SAE-AMS3218

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2016-171)

Review activities:
Army - AM, AT, CR4, MI
Navy - AS, MC, OS, SA, SH
Air Force - 19, 99

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